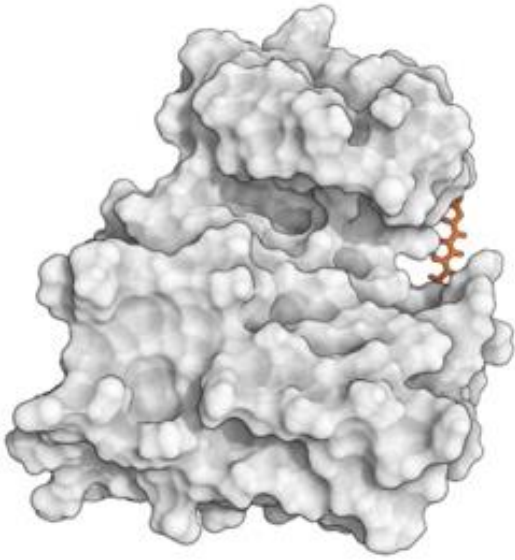


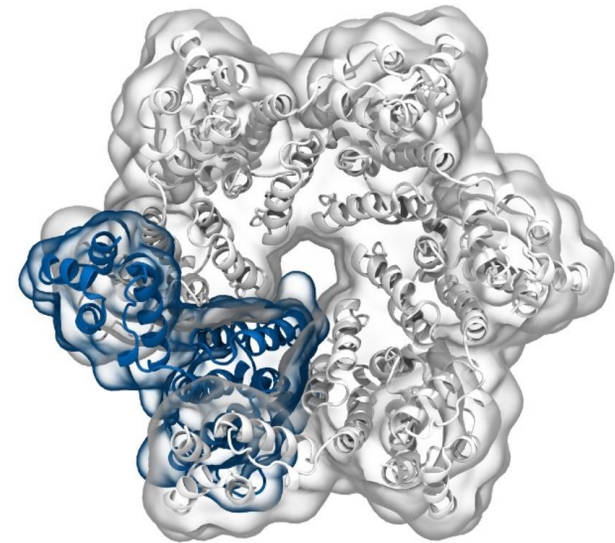
Simulation of Biomolecules

Docking



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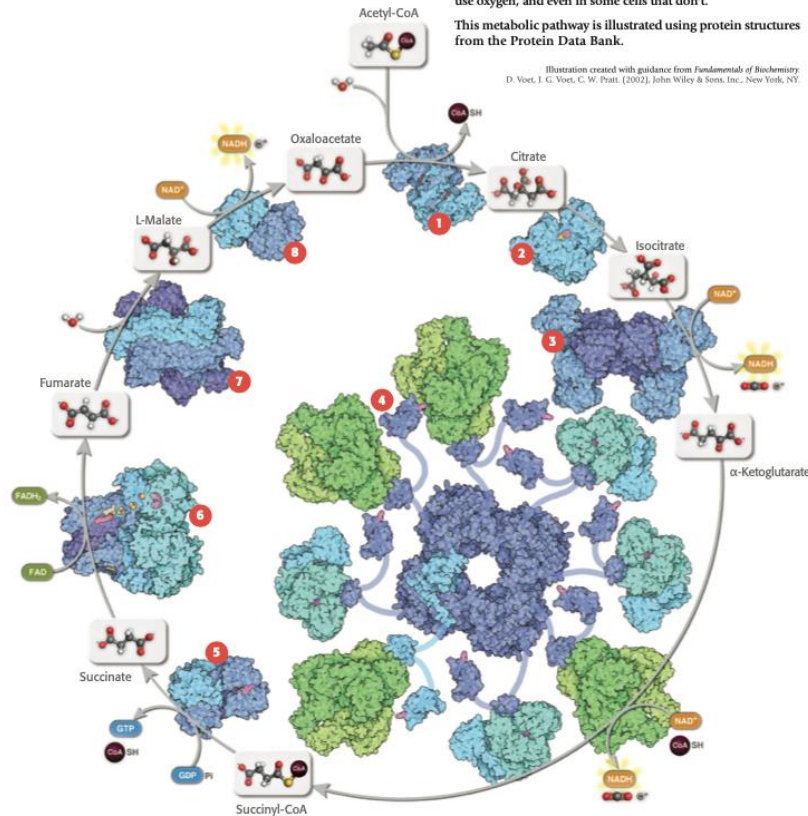
Life is built on protein and small molecule interactions

The Structures of the Citric Acid Cycle

Also known as the Krebs cycle or the tricarboxylic acid cycle, the *citric acid cycle* is at the center of cellular metabolism. It plays a starring role in both the process of energy production and biosynthesis. The cycle finishes the sugar-breaking job started in glycolysis and fuels the production of ATP in the process. It is also a central hub in biosynthetic reactions, providing intermediates that are used to build amino acids and other molecules. Citric acid cycle enzymes are found in all cells that use oxygen, and even in some cells that don't.

This metabolic pathway is illustrated using protein structures from the Protein Data Bank.

Illustration created with guidance from Fundamentals of Biochemistry
D. Voet, J. G. Voet, C. W. Pratt. [2002], John Wiley & Sons, Inc., New York, NY



Eight Reactions

The eight reactions of the citric acid cycle use the small molecule *oxaloacetate* as a catalyst. The cycle starts by addition of an acetyl group to oxaloacetate, then, over the course of eight steps, the acetyl group is completely broken apart, finally restoring the oxaloacetate molecule for another round.



Small molecules are:

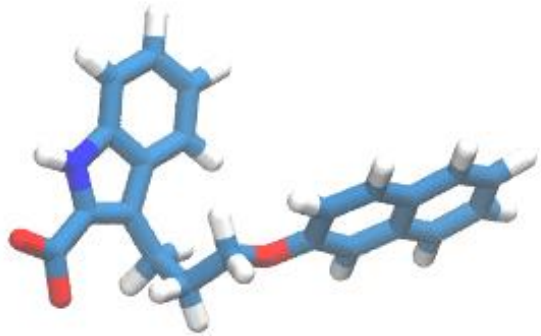
- Substrates of enzymes
- Inhibitors or activators
- Co-factors

And play an important role in life.
Accurate interaction prediction is essential.

What is docking?

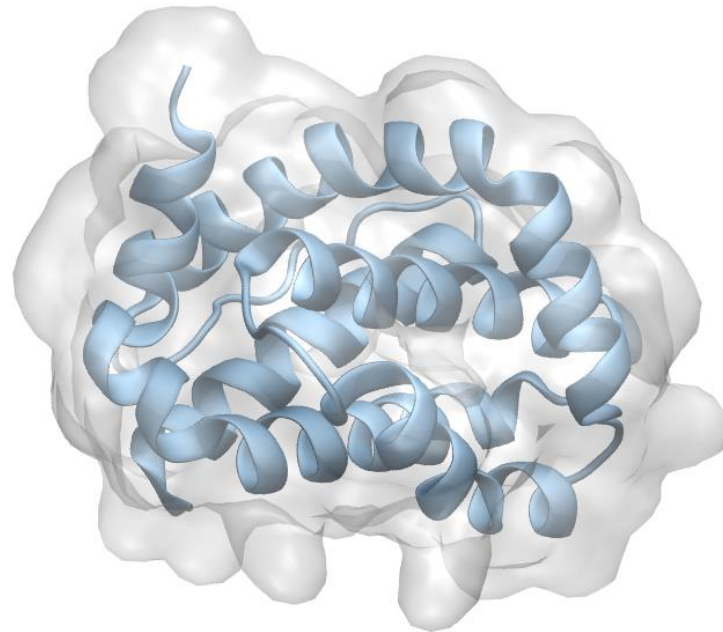
The process of predicting a stable 3D geometry of an interacting pair of molecules – a **binding mode/pose**.

Ligand



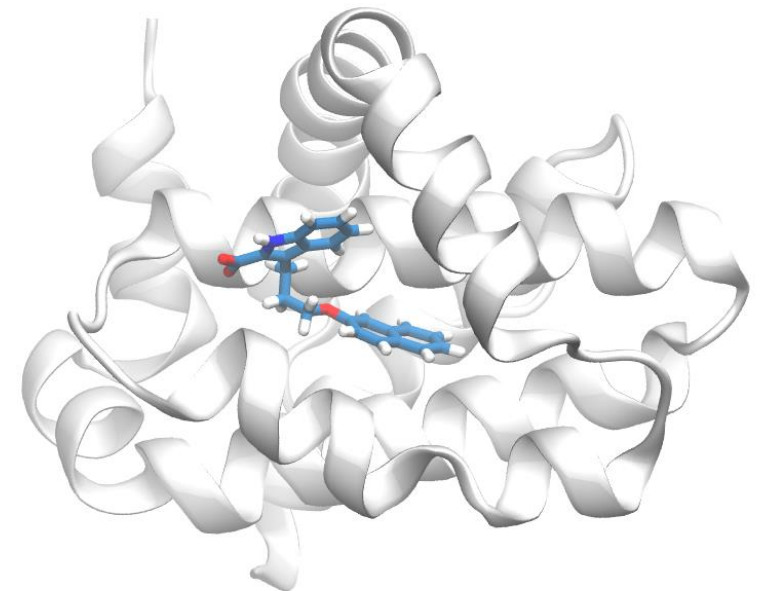
+

Receptor



=

Binding pose



Nomenclature

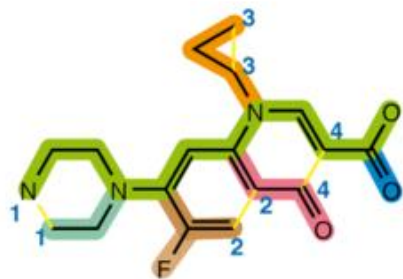
Ligand: Structure (usually a small molecule) that binds to the binding site.

Receptor: Structure (usually a protein) that contains the active binding site.

Binding site: Set of aminoacids (residues) that physically interact with the ligand (usually within 6 Ångstroms).

Typical workflow

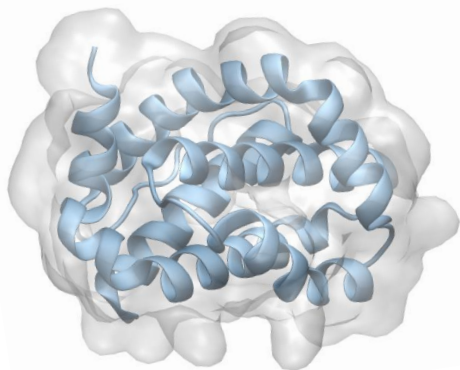
1-D or 2-D ligand structure



N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

+

Protein structure



Find Pocket

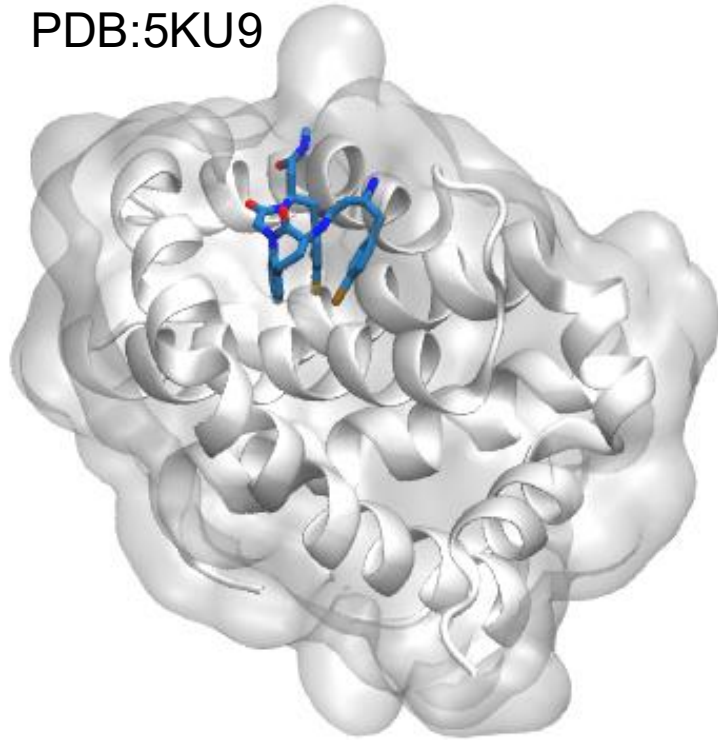
Search

Scoring Function

Docking Assessment

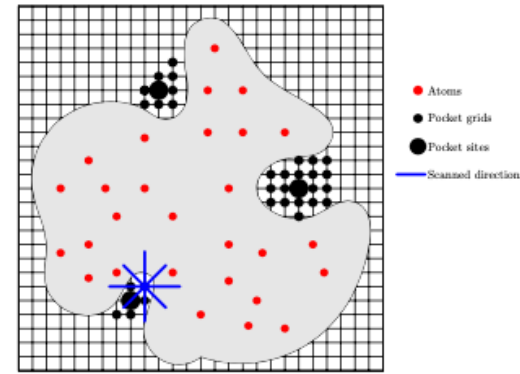
Finding a pocket

PDB:5KU9

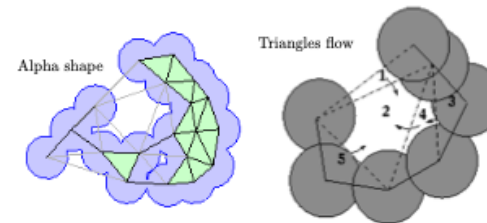


Using a reference atomic
structure with an existing
molecule bound

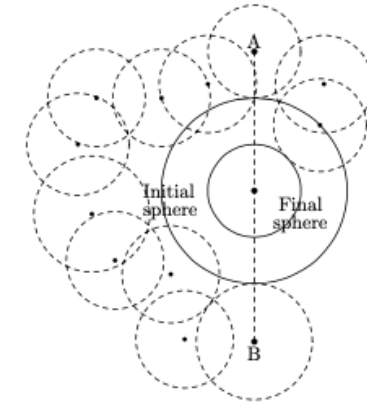
a. POCKET, LIGSITE, LIGSITE^{csc}



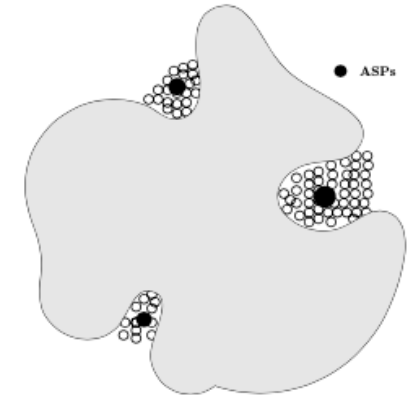
c. CAST



b. SURFNET

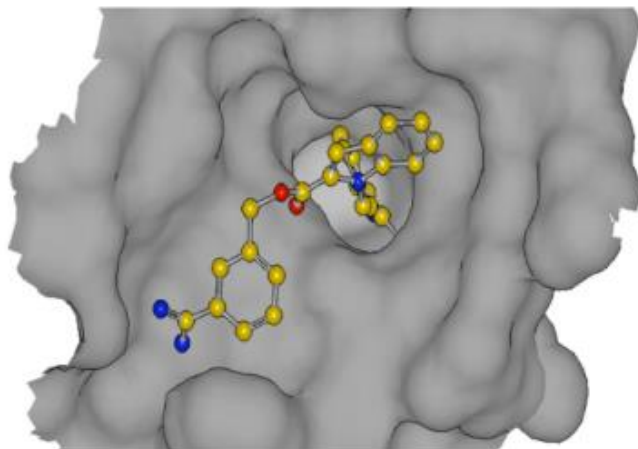


d. PASS

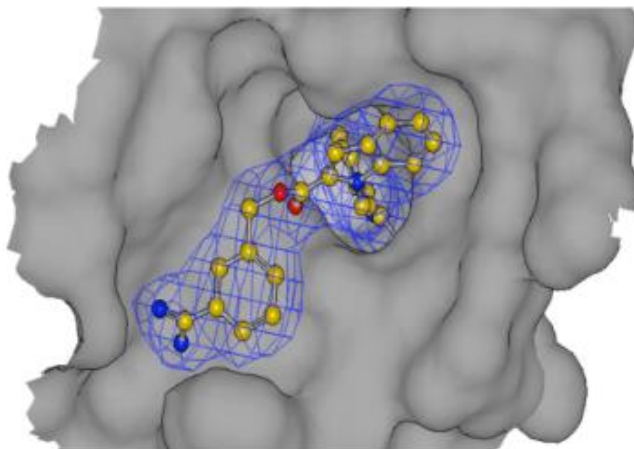


Using a pocket finding algorithm

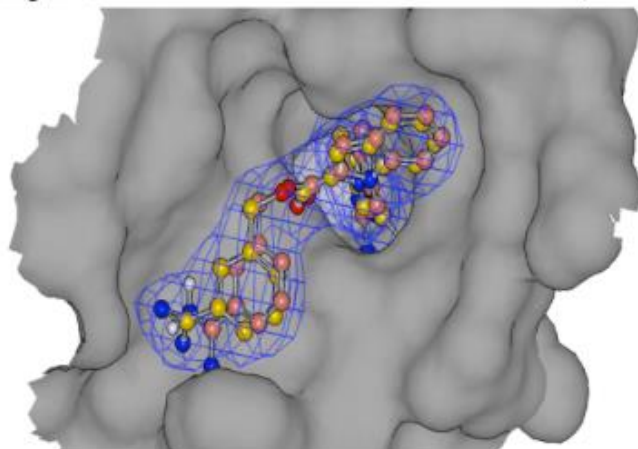
Shape based methods



Find Bound Ligand



Identify Shape Constraint



Use Shape Constraint to Optimize Overlay

With an existing ligand it is possible to match the shape and optimize the overlay

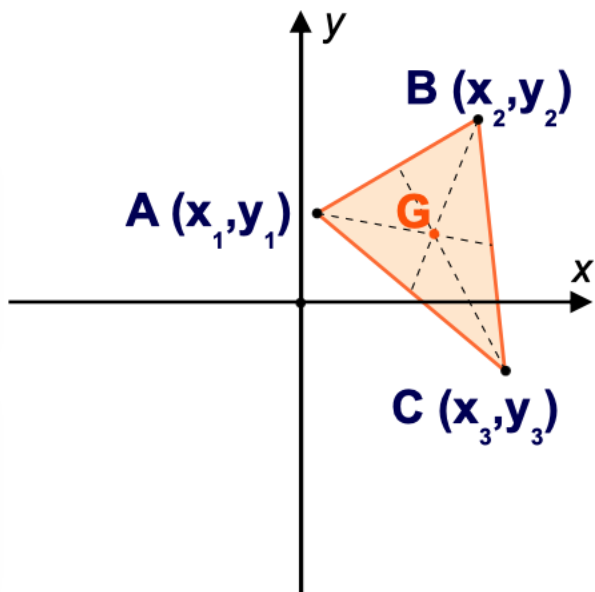
Fast and robust

Ligand changes are not taken into account.

Ligands need to be of similar size.

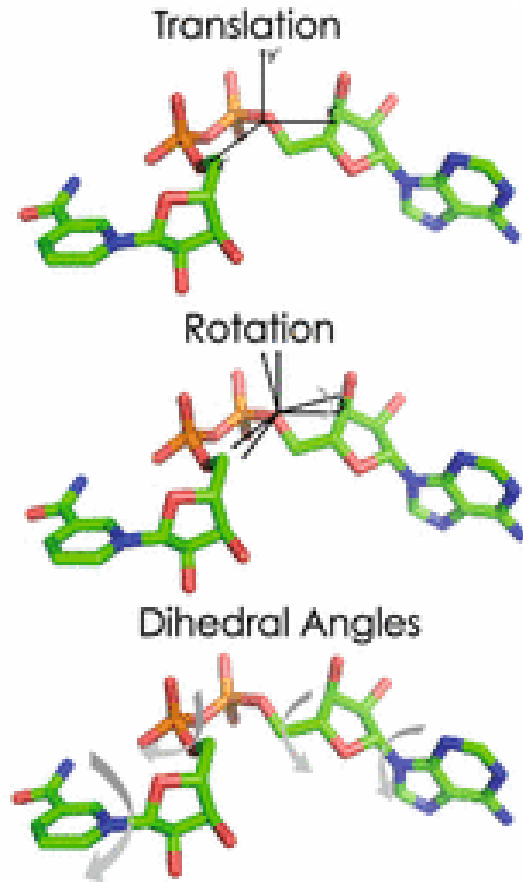
Finding the docking grid area

Often you have a ligand template or binding site residue to help with designing the docking grid

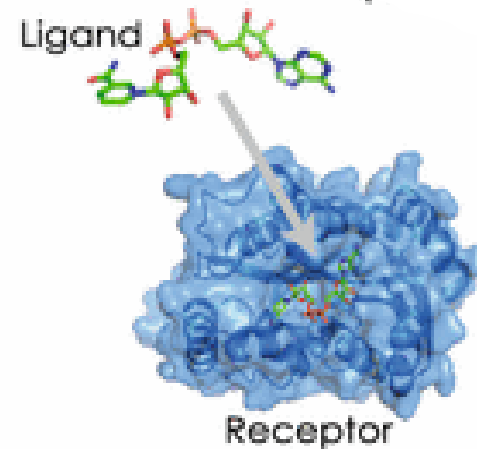


Use centre of geometry of a molecule from structure

Genetic algorithm for ligand conformers

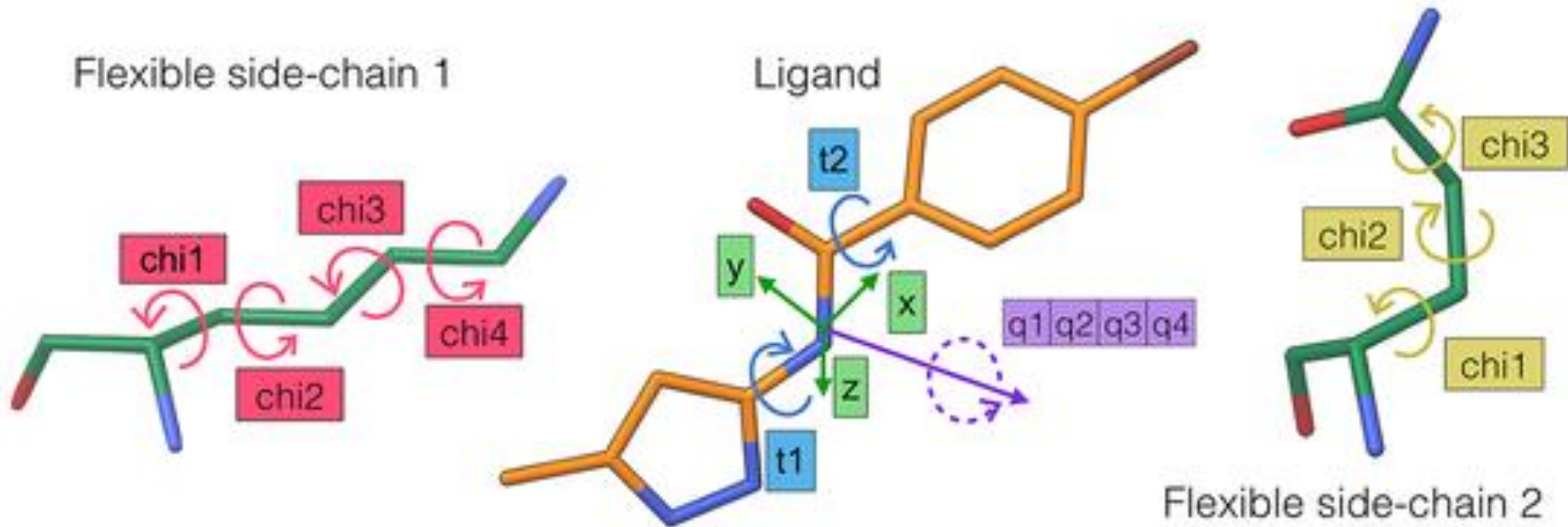


Testing different arrangements of the ligand in the rigid binding site of the protein

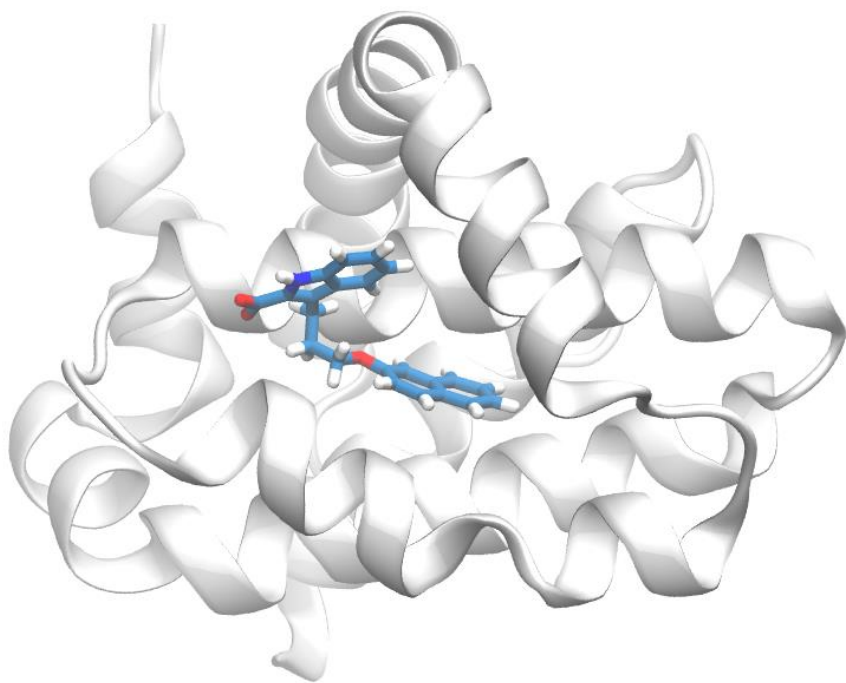


Allowing protein and ligand flexibility is often better

		Genome															
		Ligand								Receptor							
Genes		Translation			Rotation				Torsion 1	Torsion 2	Flexible side-chain 1				Flexible side-chain 2		
Variables		x	y	z	q1	q2	q3	q4	t1	t2	chi1	chi2	chi3	chi4	chi1	chi2	chi3



Flexibility increases compute time



$$N = T \cdot 360 / i$$

N: number of conformations

T: number of rotatable bonds

i: incremental degrees

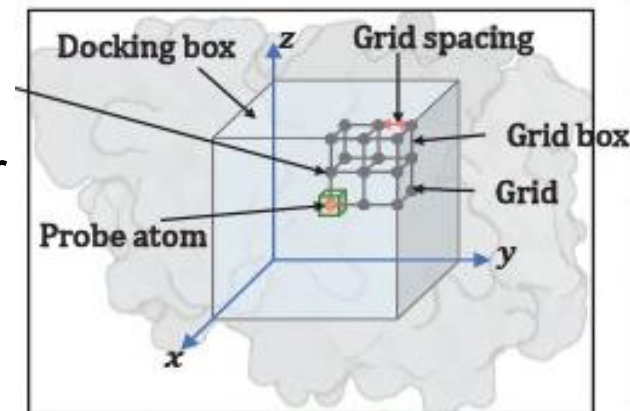
Typical drug molecule

10 rotatable bonds

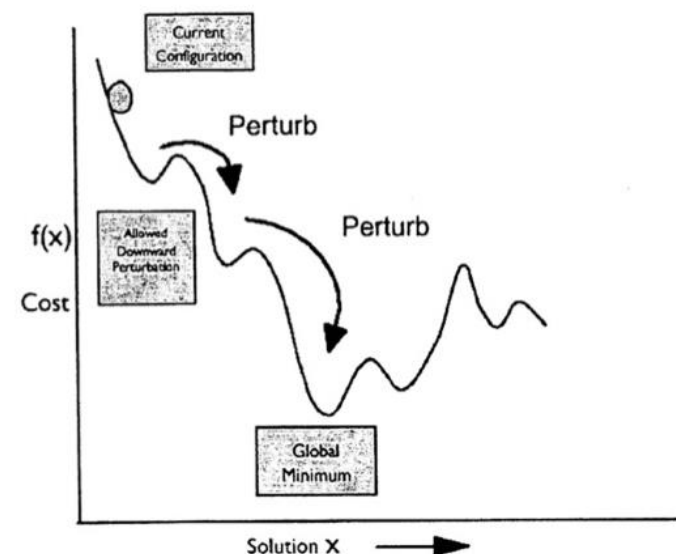
30° increments (discrete)

10¹² plausible conformations!

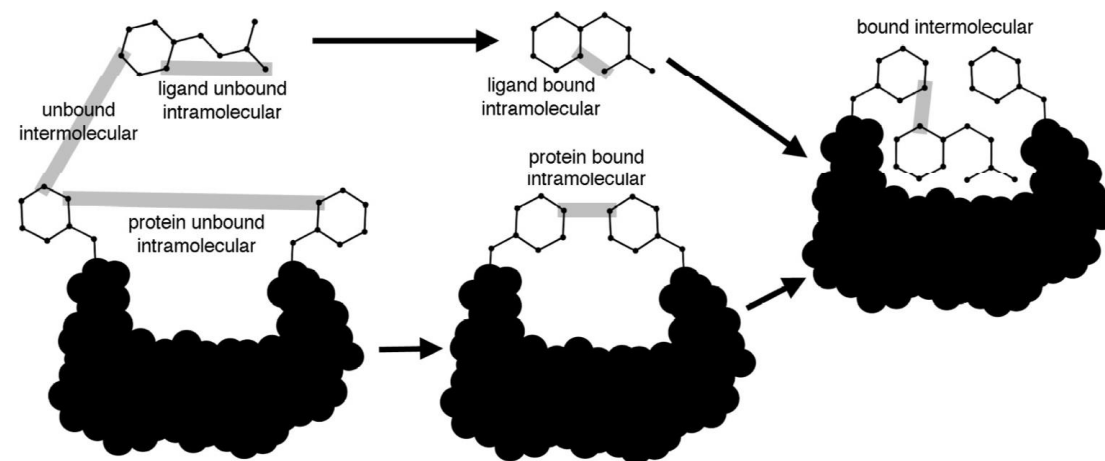
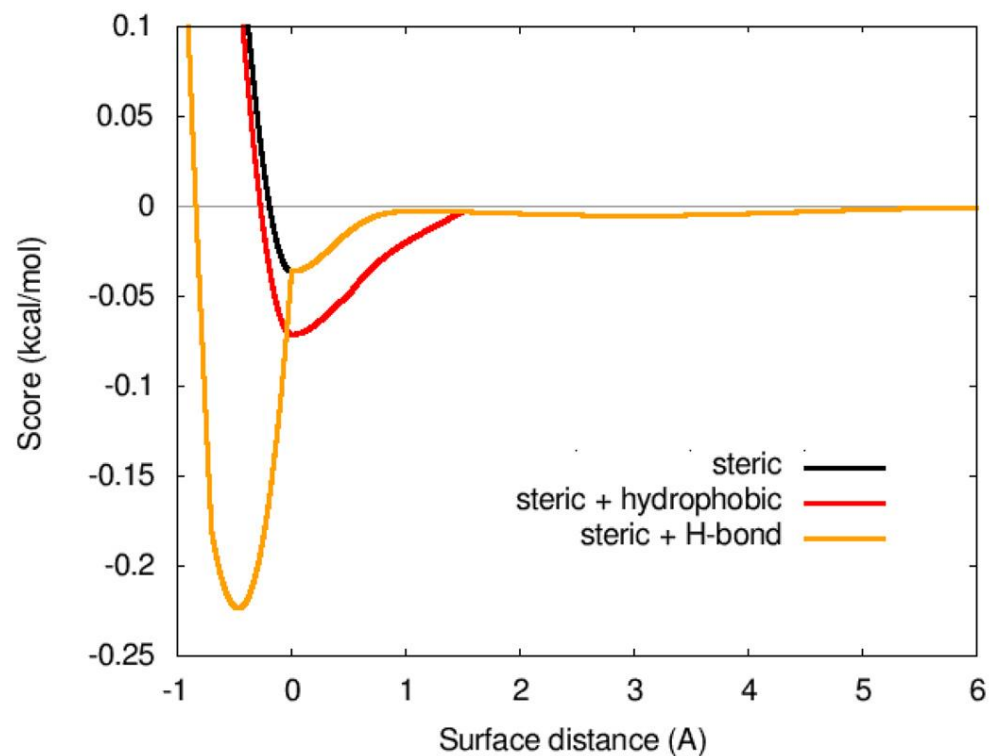
Pre-calculate
interaction terms for
all atom types on a
grid:



Simulated annealing

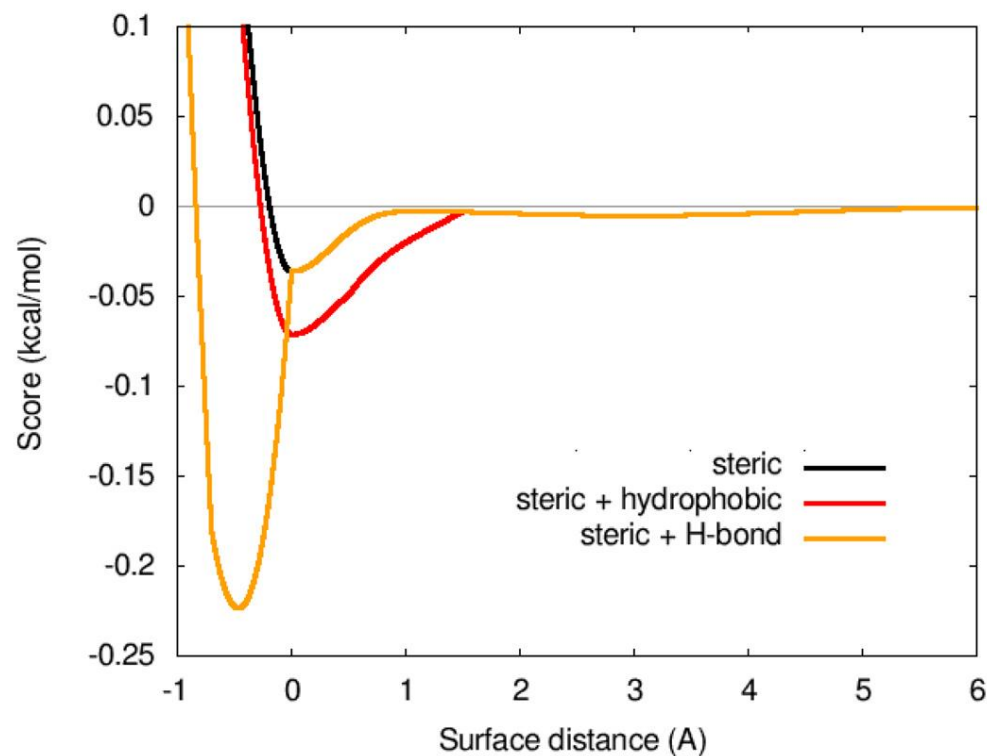


Scoring functions



Scoring functions can be used beyond shape optimization to optimize ligand and protein interactions

Scoring functions



Scoring functions can be used beyond shape optimization to optimize ligand and protein interactions

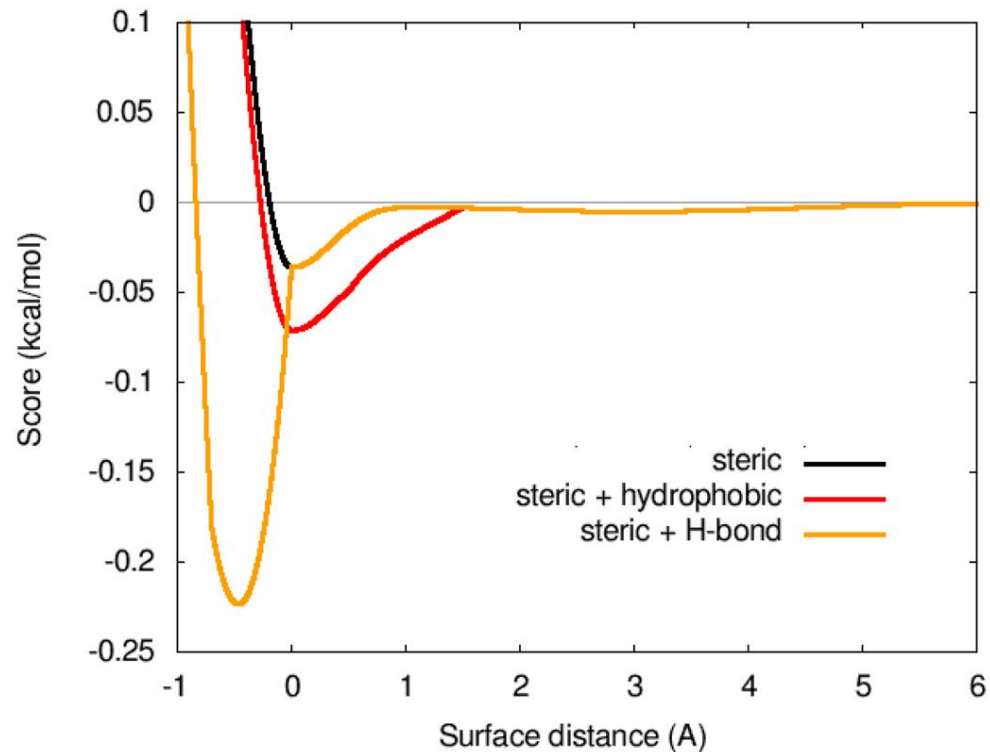
$$\Delta G = (V_{bonded}^{L-L} - V_{unbonded}^{L-L}) + (V_{bonded}^{R-R} - V_{unbonded}^{R-R}) + (V_{bonded}^{R-L} - V_{unbonded}^{R-L} + \Delta G_{conf})$$

$$V = W_{vdw} \sum_{i,j} \left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right) + W_{hbond} \sum_{i,j} E(t) \left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}} \right) + W_{elec} \sum_{i,j} \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} + W_{sol} \sum_{i,j} (S_i V_j + S_j V_i) e^{\frac{-r_{ij}^2}{2\sigma^2}}$$

V = volume of surrounding atoms shielding from solvent

$$\Delta G_{conf} = W_{conf} N_{tors}$$

Scoring functions



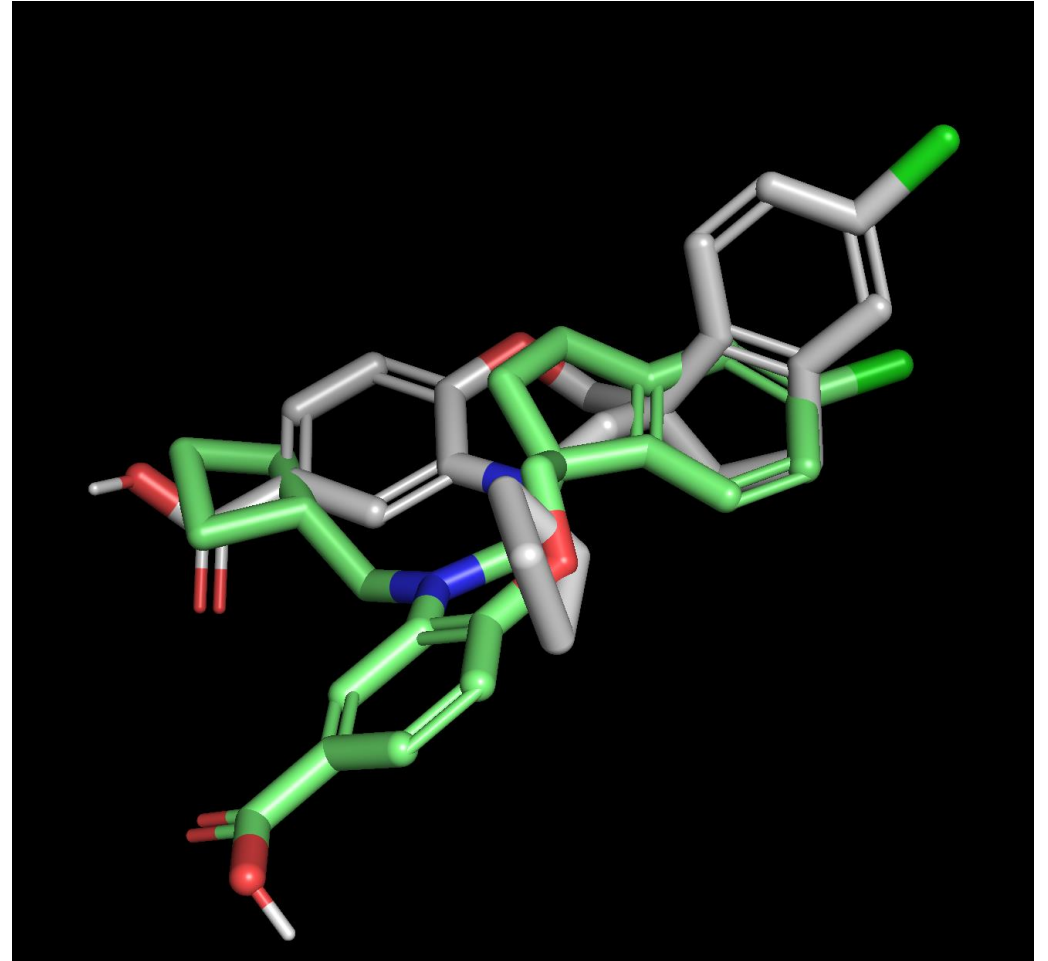
Scoring functions can be used beyond shape optimization to optimize ligand and protein interactions

Weight	Term
-0.0356	gauss ₁
-0.00516	gauss ₂
0.840	repulsion
-0.0351	hydrophobic
-0.587	hydrogen bonding
0.0585	N_{rot}

Weights are optimised based on experimental measurements (semi-empirical)

Typical docking output generates multiple poses

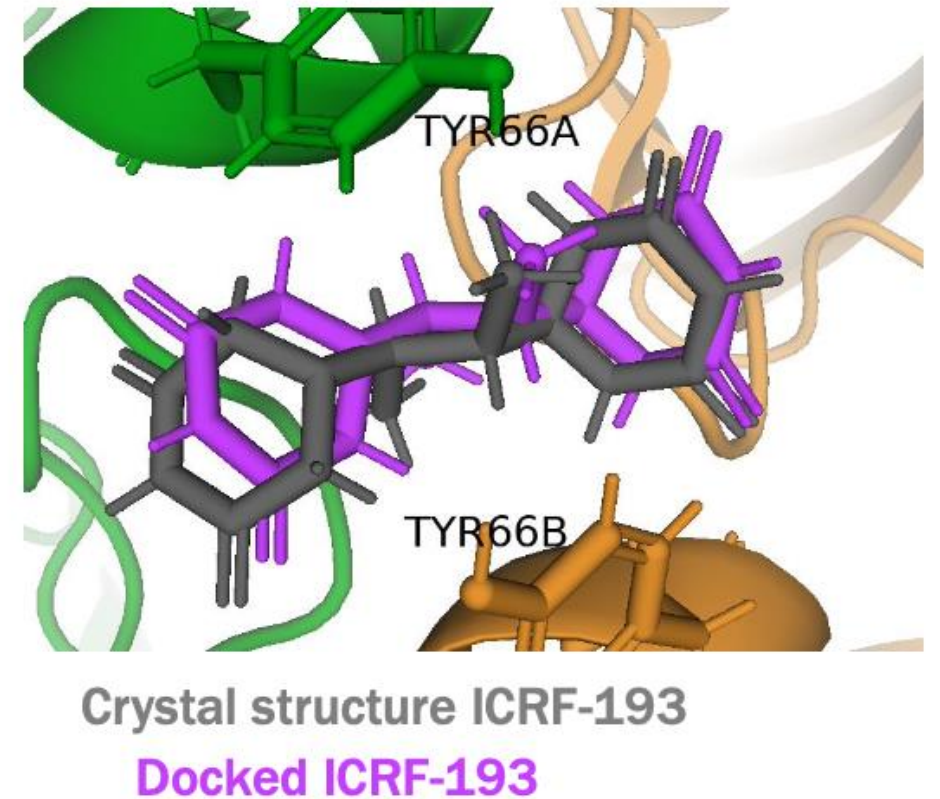
mode	affinity	dist from best mode	
	(kcal/mol)	rmsd l.b.	rmsd u.b.
-----+-----+-----+-----			
1	-8.36	0	0
2	-8.08	2.899	6.789
3	-7.985	3.643	7.852
4	-7.914	3.415	5.21
5	-7.765	2.167	2.826



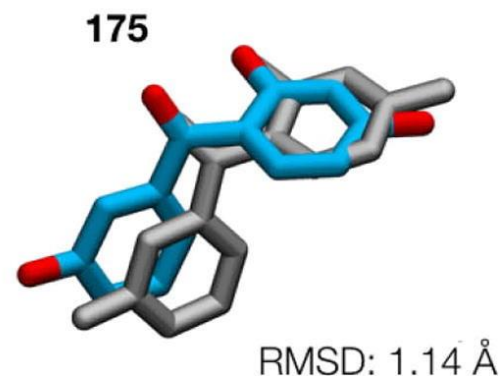
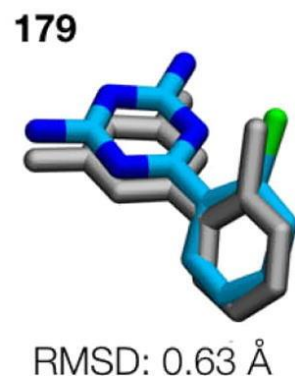
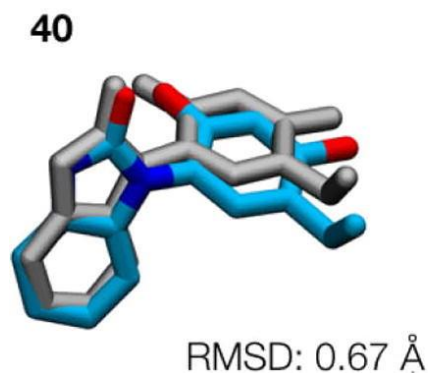
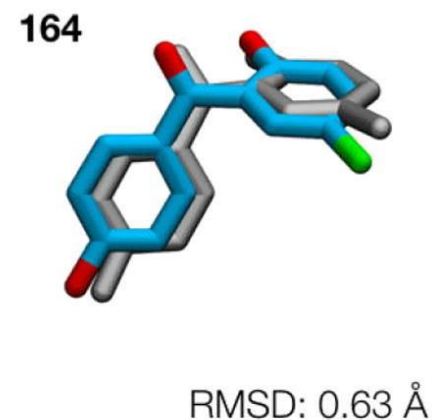
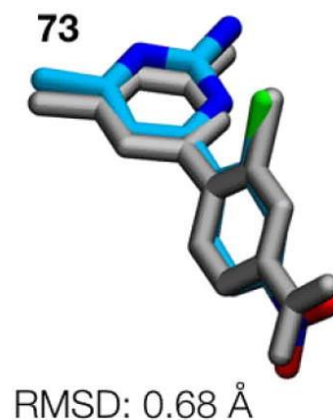
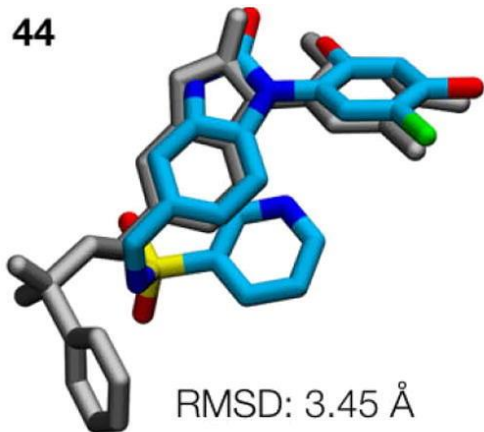
How good is my docking pose?

- Let a ligand with N atoms, an experimental reference structure x_0 , and a predicted pose x_1 .
- The *pose accuracy* is quantified with the Root Mean Square Deviation (RMSD):

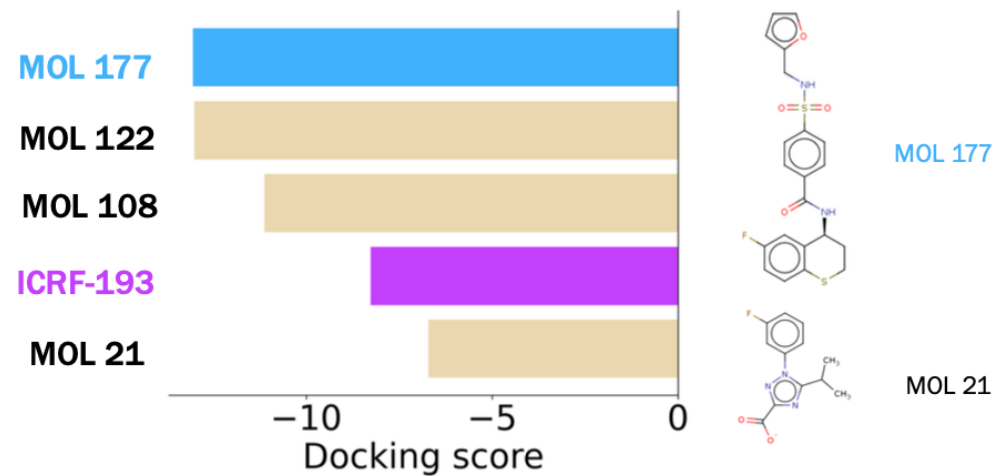
$$RMSD = \sqrt{\frac{1}{N} \sum_{i=0}^N (x_1 - x_0)^2}$$



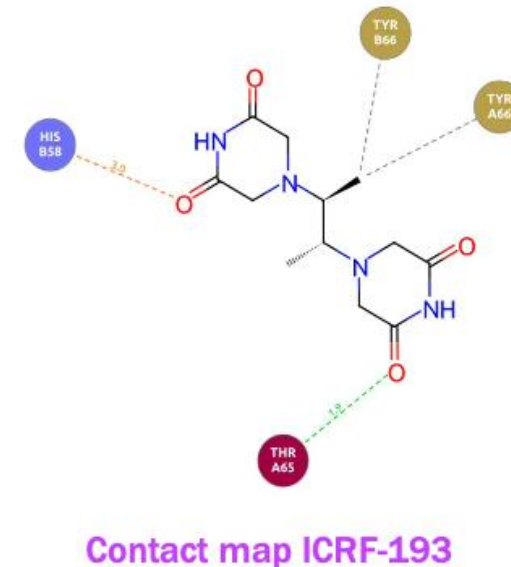
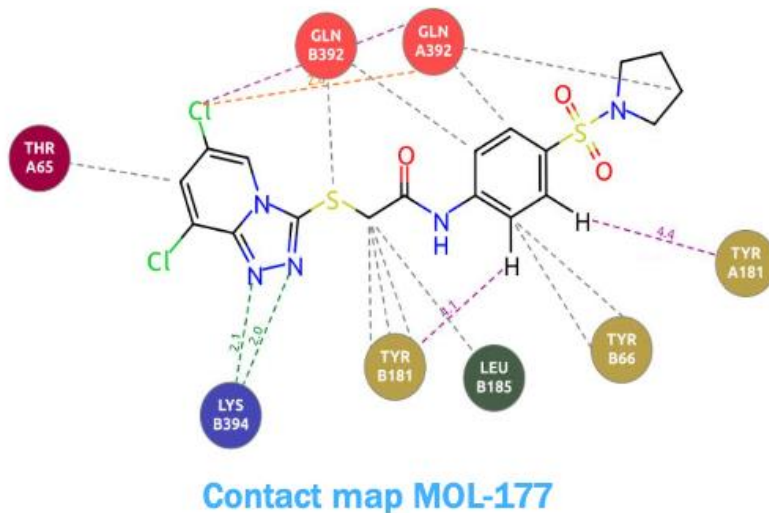
Examples of docking poses



How good is my docking pose?

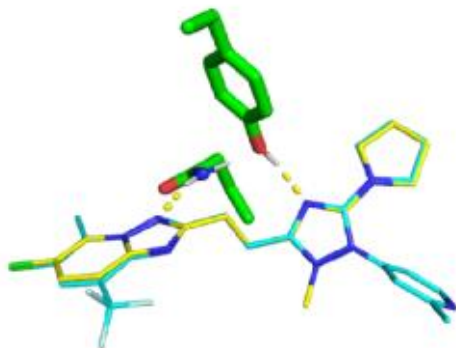


- The *score* gives a metric on how well the ligand can be expected to bind to the pocket
- Interaction diagrams show favourable interactions and clashes

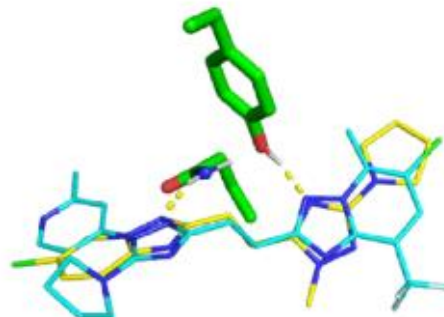


Template docking and cross docking improves docking

(a)



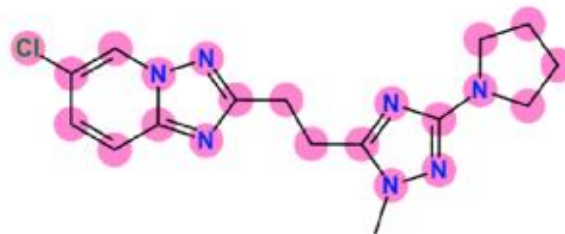
(b)



(c) Ligand compound 40:



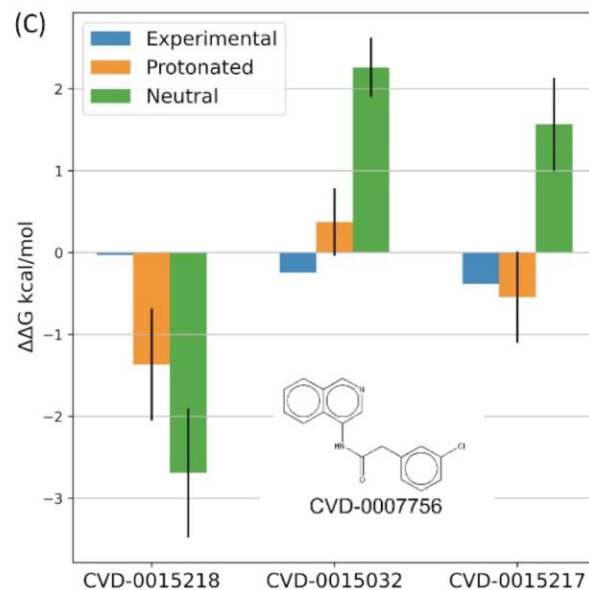
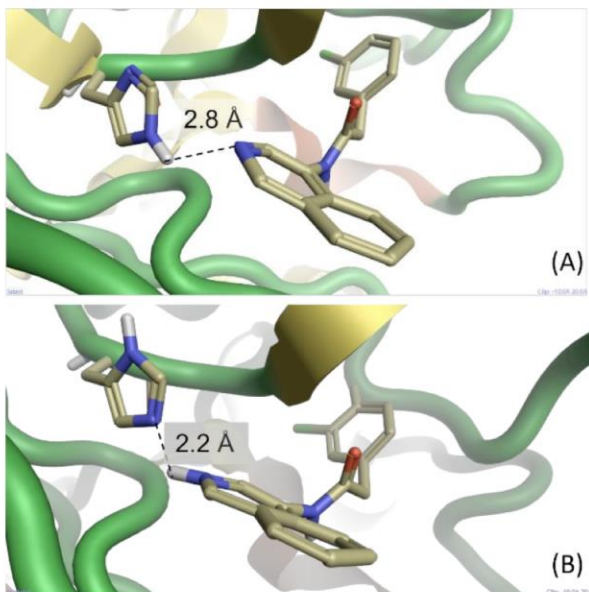
(d) Template compound PDB code 5sej:



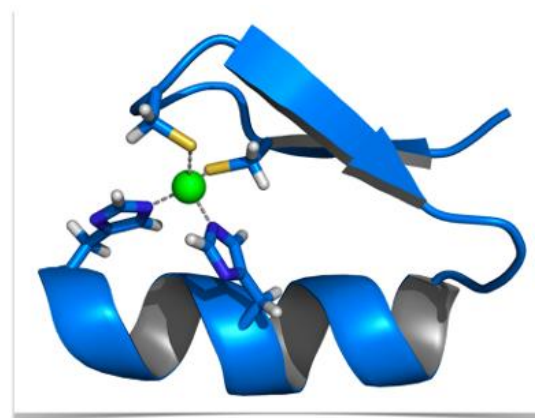
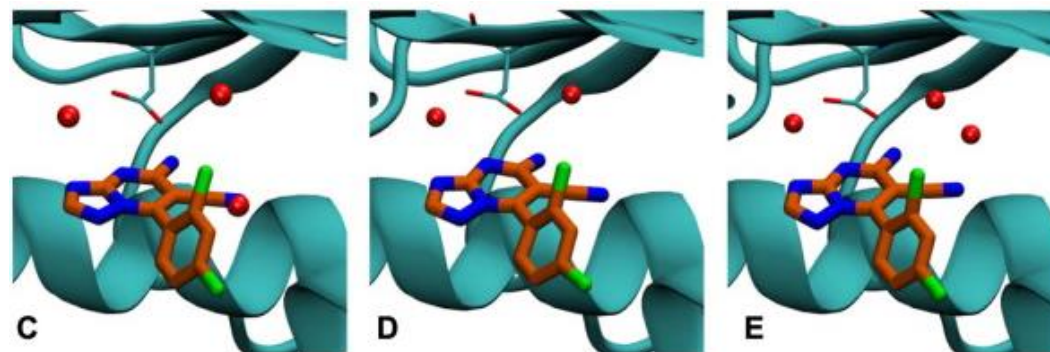
- Dock the same ligand into multiple protein structures (X-ray, MD)
- Generate multiple ligand conformers and dock into multiple structures
- Use existing ligand data as a template or guide e.g. through Maximum common substructure

Things to worry about

pKa of ligands and binding site
protonation need to be considered



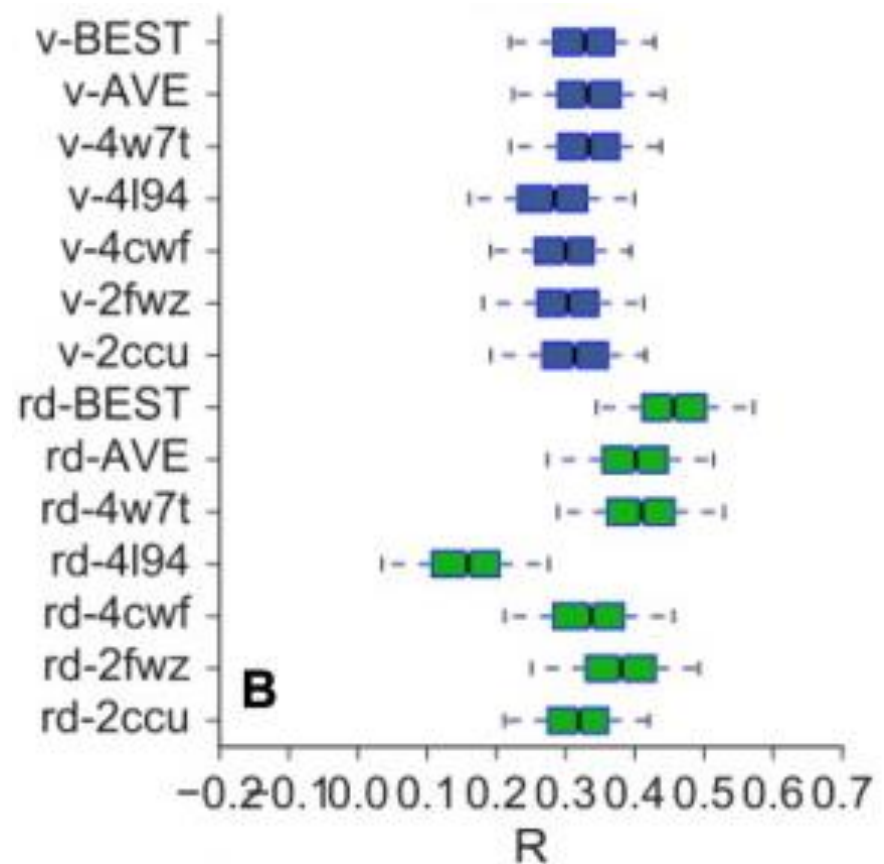
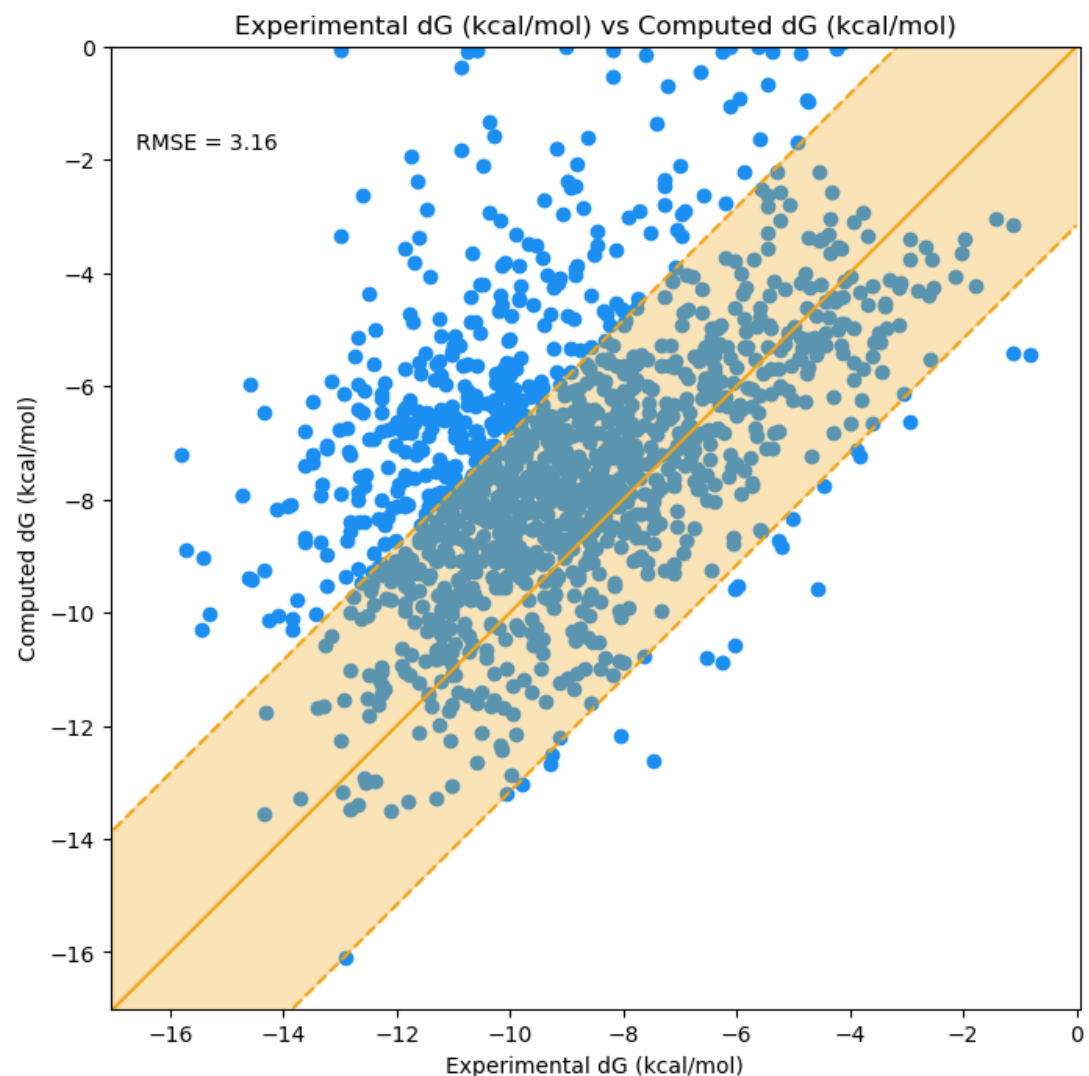
Structural waters are important



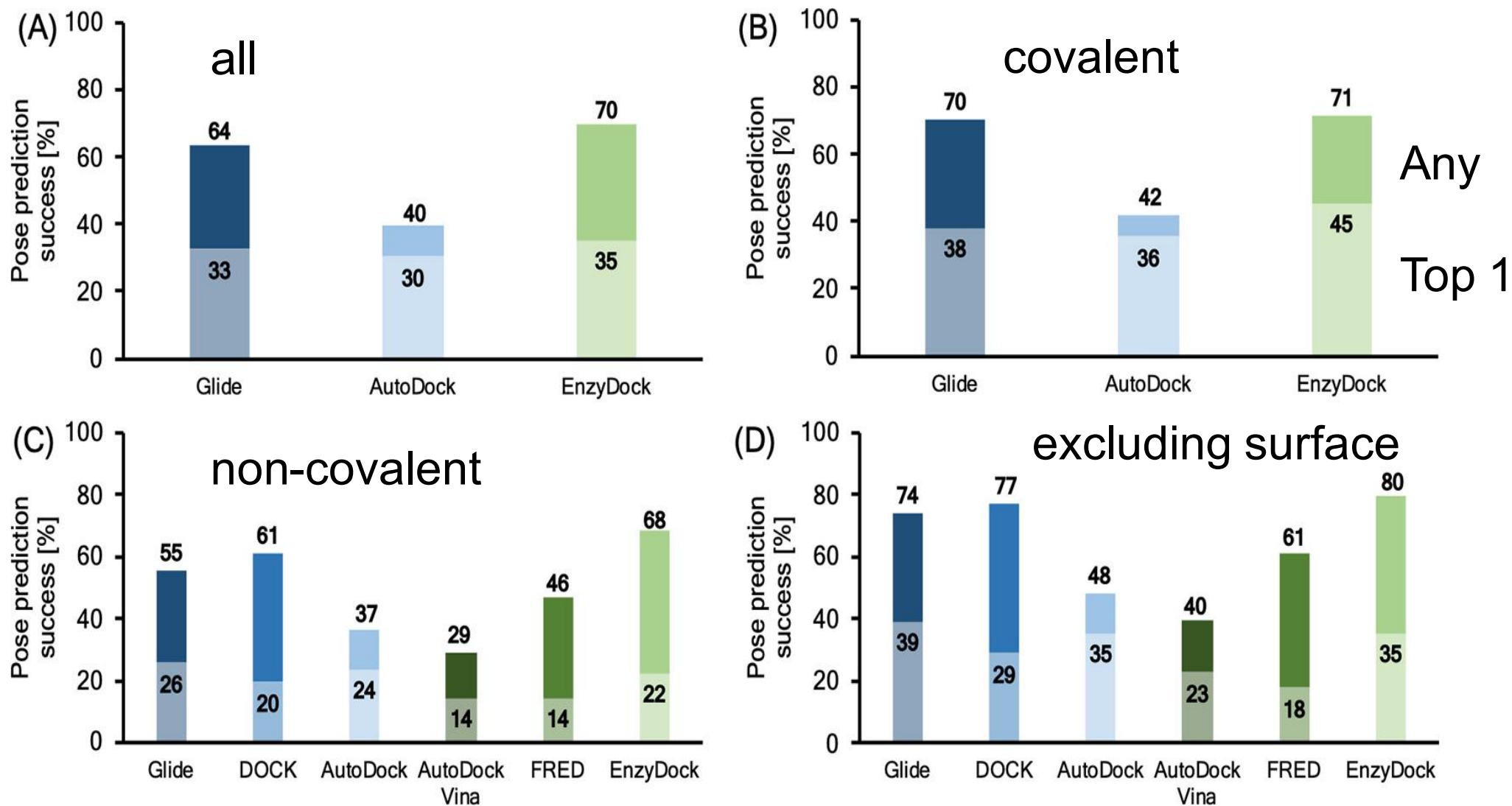
Co-factors such as
ions and other
molecules are
important

Comparing against experimental ΔG

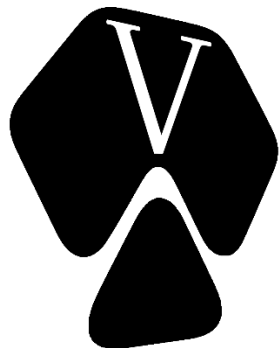
Bulk assessment of correlation coefficient



Recent docking benchmark



What tools exist for molecular docking?



Autodock Vina



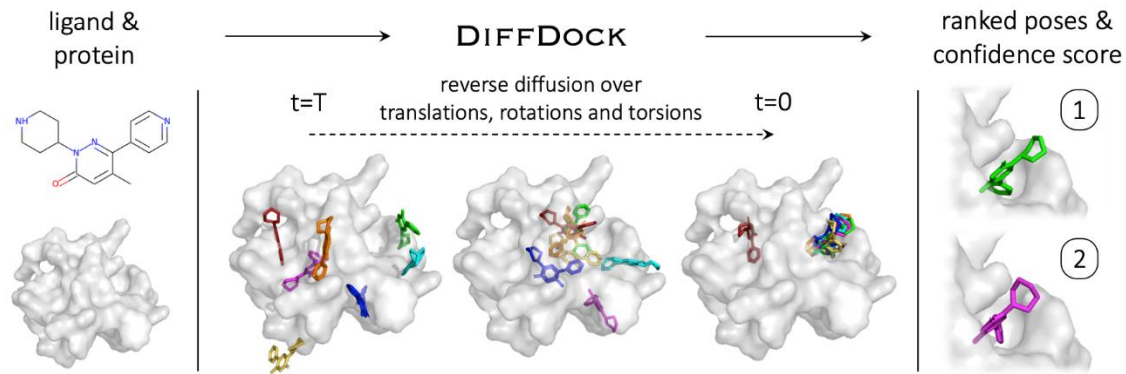
OPEN-SOURCE



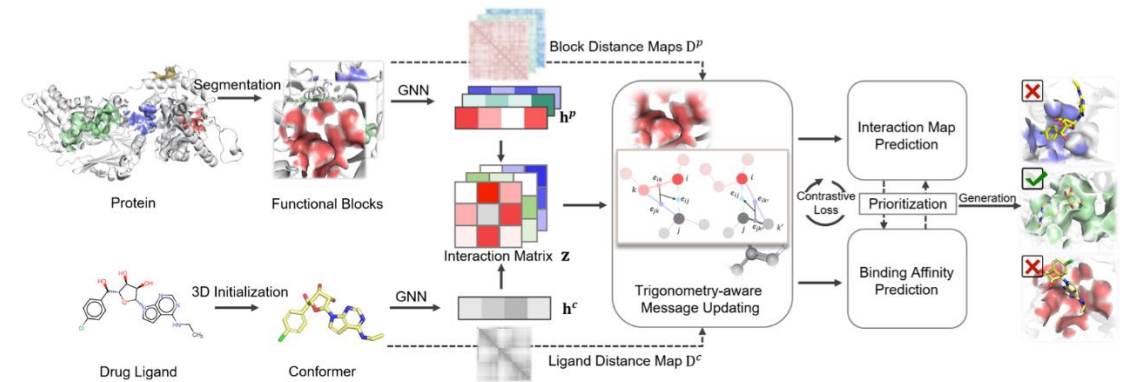
And others...

ML-based docking

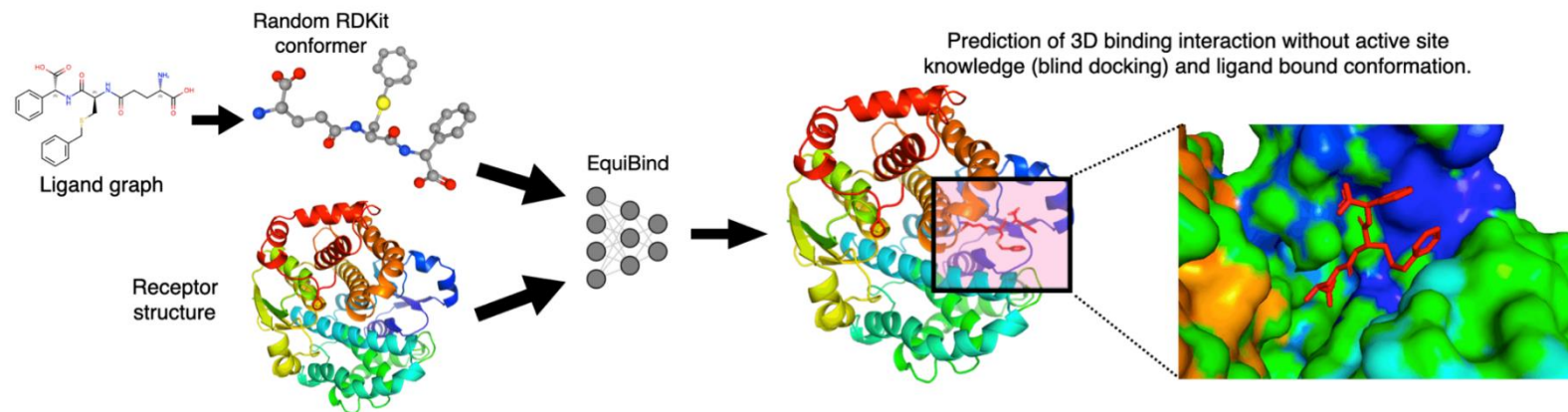
Diffdock



TankBind



Equibind



Gnina

Deepdock

...

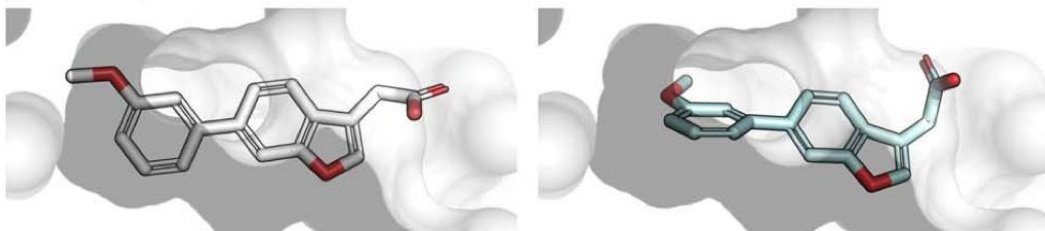
Evaluating the binding mode/pose of ML tools



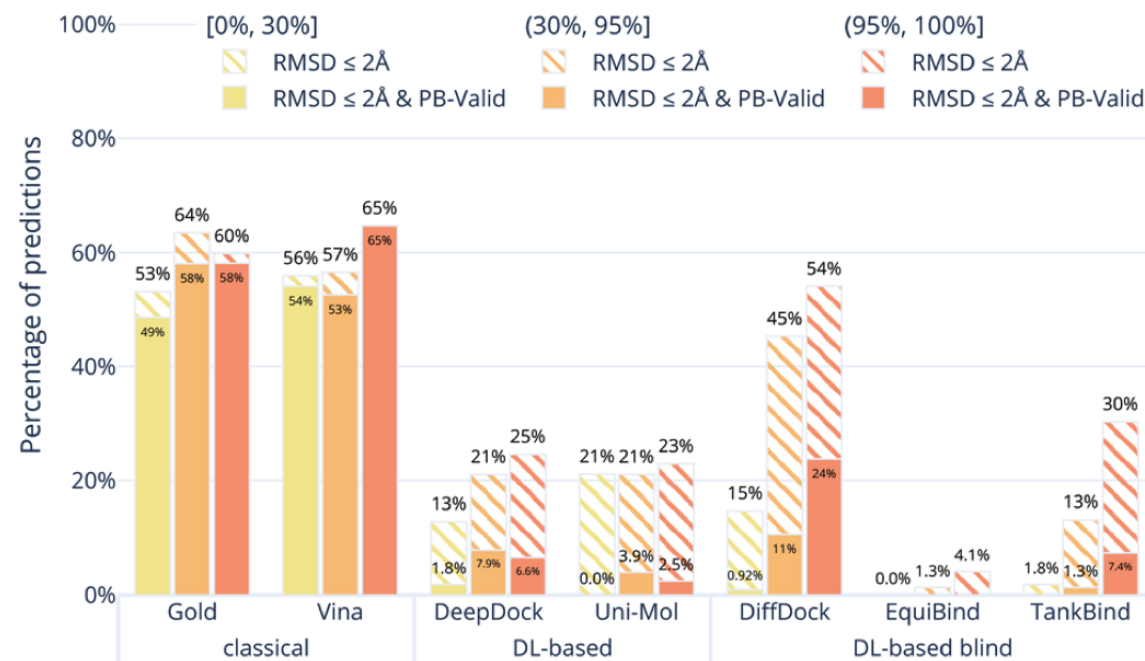
(f) Double bond not flat. TankBind prediction for ligand DBQ of protein-ligand complex 1U4D. RMSD 1.7 Å.



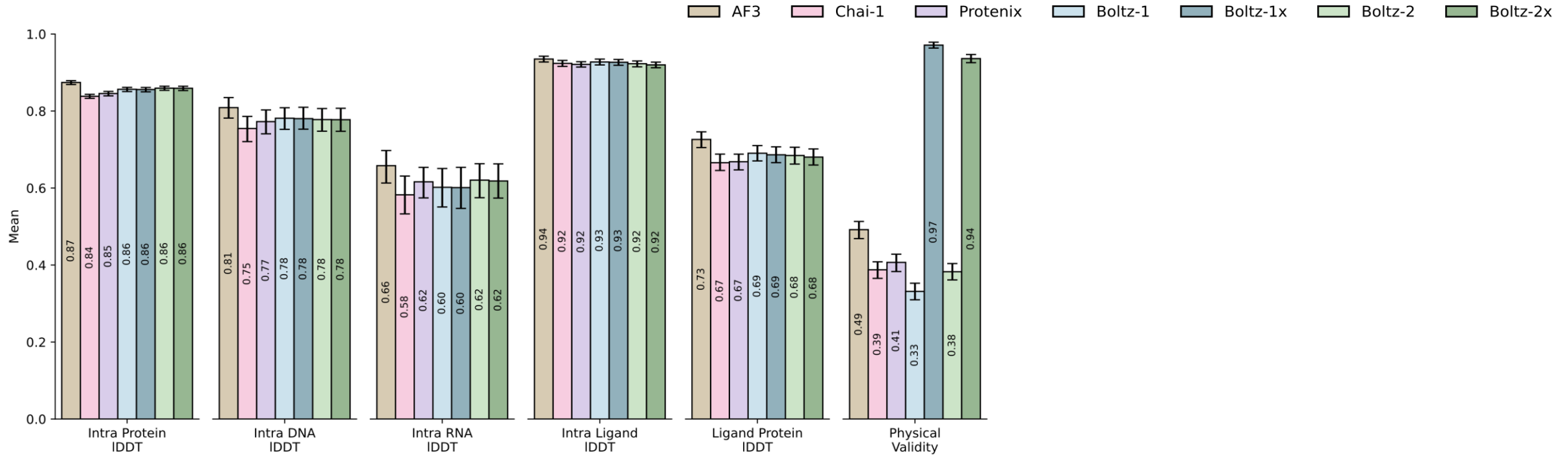
(g) Energy ratio too high. AutoDock Vina prediction for ligand IFM of protein-ligand complex 7LOU. RMSD 1.9 Å.



(h) Clash with protein. DiffDock prediction for ligand XQ1 of protein-ligand complex 7L7C. RMSD 1.6 Å.



Boltz-2

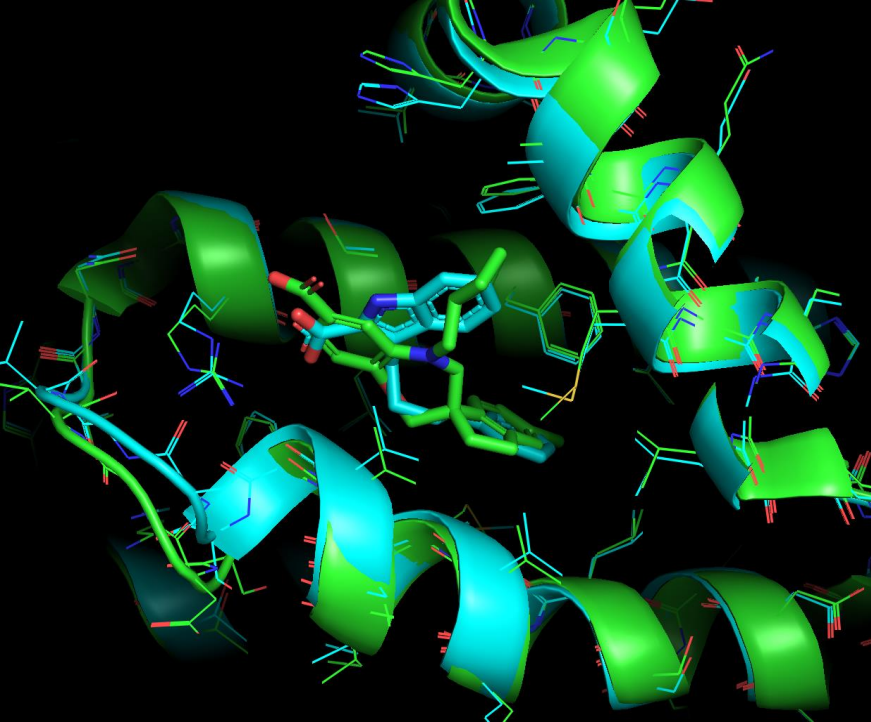


Open source model for structure and binding affinity prediction, approaching AlphaFold3 accuracy.

Training data from PDB, BindingDB, ChEMBL and PubChem.

<https://github.com/jwohlwend/boltz>

Boltz-2



```
version: 1 # Optional, defaults to 1
sequences:
  - protein:
      id: A
      sequence: GSDELYRQSLEIISRYLREQATGAKDTKPMGRSGATSRKALETLLRV
GDGVQRNHETAFQGMLRKLDIKNEDDVKSLSRVMIHVFSDGVTNWGRIVTLISFGAFVAKHLK
TINQESCIEPLAESITDVLVRTKRDWLVKQRGWDGFVEFFHVEDLEGG
  - ligand:
      id: B
      smiles: O=C([O-])c1[nH]c2cccc2c1CCC0c1cccc1
```

`boltz predict --use_msa_server --output_format=pdb --use_potentials --diffusion_samples=1 input/`

<https://github.com/jwohlwend/boltz>